1. Explain Public IP?
2. What is Private IP?
3. What is IPV4?
4. Explain Networking between two or more computers.

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Aim: Demonstrate the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

Understand the concept of RDP for remotely controlling another computer from our system, learn other remote desktop solutions, and learn the data security and data sharing concepts.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirement

- Operating systems: Windows 10 1703 or later.
- CPU: 1 GHz or faster processor.
- RAM: 1024 MB.
- Hard drive: 100 MB or more.
- Video: DirectX 9 or later with WDDM 1.0 driver.

THEORY:

STEPS FOR PRACTICAL:

- **What is the Remote Desktop Protocol (RDP)?**

  The Remote Desktop Protocol (RDP) is a protocol, or technical standard, for using a desktop computer remotely. Remote desktop software can use several different protocols, including RDP, Independent Computing Architecture (ICA), and virtual network computing (VNC), but RDP is the most used protocol. RDP was initially released by Microsoft and is available for most Windows operating systems, but it can be used with Mac operating systems too.

- **What does 'remote desktop' mean?**

  A remote desktop is the ability to connect with and use a faraway desktop computer from a separate computer. Remote desktop users can access their desktop, open and edit files, and use applications as if they were sitting at their desktop computer. Employees often use

remote desktop software to access their work computers when they are traveling or working from home.

- Remote desktop access is very different from cloud computing, even though both allow employees to work remotely. In cloud computing, users access files and applications that are stored in the cloud — specifically, in cloud servers. In contrast, when using remote desktop software, users are accessing their physical desktop computer, and can only use files and applications saved locally on that desktop. Cloud computing is sometimes easier to use and more efficient to implement for remote workforces, but many companies have not migrated to the cloud, or cannot for security or regulatory reasons.
- **How does RDP work?**

  Think of a remote-controlled drone or toy car. The user presses buttons and steers the drone or car from afar, and their commands are transmitted to the vehicle. Using RDP is somewhat like that: the user's mouse movements and keystrokes are transmitted to their desktop computer remotely, but over the Internet instead of over radio waves. The user's desktop is displayed on the computer they are connecting to, just as if they were sitting in front of it.
- The RDP protocol opens a dedicated network channel for sending data back and forth between the connected machines (the remote desktop and the computer currently in use). It always uses network port 3389 for this purpose. Mouse movements, keystrokes, the desktop display, and all other necessary data are sent over this channel via TCP/IP, which is the transport protocol used for most types of Internet traffic. RDP also encrypts all data so that connections over the public Internet are more secure.
- Because keyboard and mouse activity have to be encrypted and transmitted over the Internet, which takes a few milliseconds, and because the desktop display has to be transmitted back to the user, slight delays often occur. For instance, if a user double-clicks on an application to open it, the "double click" may not take place for a few milliseconds as the user's action is transmitted to the desktop before being carried out. Then, when the application opens, there may be another short delay while the display is transmitted back to the user.
- **What are the pros and cons of using RDP?**

There are several advantages to the RDP. One advantage is that it does not require a VPN. It also keeps data stored securely on the user's desktop, instead of storing it on cloud servers or the user's unsecured personal devices. Furthermore, RDP enables companies with a legacy on-premises IT setup to allow their employees to work from home.

However, RDP may cause users to experience lag, especially if their local Internet connection is slow. This can frustrate remote employees and cut down on their productivity. RDP also has some serious security vulnerabilities that leave it open to cyber attacks. (Learn more about RDP security.) **Steps to Secure Your RDP**

1. **Rigorous Patch Management:**

   Continually update both the operating system and applications. This includes subscribing to updated services and conducting regular audits to ensure all systems are up-to-date.

2. **Multi-Factor Authentication (MFA):**

   Beyond two-factor, consider implementing multi-factor authentication. This approach combines something you know (password), something you have (security token), and something you are (biometrics), significantly increasing security.

3. **Restrict Privileged Access:**

   Implement the principle of least privilege by ensuring users have only the access necessary for their role. Review and adjust these permissions regularly to adapt to changing roles and responsibilities.

4. **Strong, Unique Passwords**

   Enforce the use of strong, unique passwords for RDP access. Employ password managers to generate and store complex passwords and mandate regular password changes.

5. **Secure RDP Files**

   Avoid storing passwords in RDP files and disable the option to save credentials as a default setting on client devices. This prevents credential theft of lost or stolen devices.

6. **Manage RDP Connection History**

   Regularly clean the RDP connection cache to remove residual information that could be used in an attack. Automate this process through scripts or group policies where possible

7. **Network Level Authentication (NLA)**

Require NLA for RDP sessions, which necessitates that the connecting user authenticate before a session is established, thus providing an additional layer of security against unauthorized access.

8. **Configure Firewalls Strictly**

   Limit RDP access through firewalls by whitelisting IP addresses. Employ additional monitoring to detect and respond to unusual access patterns or unauthorized remote access or attempts.

9. **Deploy RDP Gateways**

   Use Remote Desktop Gateway servers as a middle layer to manage and secure RDP traffic entering the corporate network from remote systems. RD Gateways uses SSL encryption, improving security over traditional RDP.

10. **Alter Default RDP Port**

- Change the default port (3389) to a less predictable number to avoid targeted attacks on this well-known port. Monitor and potentially rotate the RDP port periodically to enhance security.

OUTPUT SCREENSHOT:

- *STICK HERE YOUR OUTPUT IMAGES DO PROPER TAGGING*

OBSERVATION:

- ***Connection establishment: TCP 3389 session, TLS negotiation (if configured), whether NLA (Network Level Authentication) is required.***
- ***Authentication prompts: NLA vs regular logon; error messages for wrong creds.***
- ***Latency & screen update behavior: responsiveness under different network conditions.***
- ***Clipboard and file transfers: whether clipboard text/images transfer, whether drives show up on the remote host.***
- ***Access control: can a non-authorized account connect? What happens when multiple clients connect to same user session (Windows behavior vs. multi-session server)?***
- ***Logging: corresponding Windows Event Log entries (Event IDs for logon/logoff, failed logons).***
- ***Effect of network controls: blocking 3389 prevents access; VPN allows access when firewall blocks public 3389.***

CONCLUSION: Hence, we understand the concept of RDP for remotely controlling another computer from our system learn other remote desktop solutions and learn the data security and data sharing concepts

- **DISCUSSION QUESTIONS:**

1. What is the Remote Desktop Protocol (RD)
2. What does 'remote desktop' mean?
3. How does RDP work?
4. What are the pros and cons of using RDP?

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